

ALFA RIZI

Junior Backend & Fullstack Developer

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Summary

Informatics Management student specialising in Backend and Fullstack Development, with a track record of delivering production-ready web applications using Java Spring Boot, PostgreSQL, and React. Proven leadership experience managing intern teams in live IT operations environments. Expanding expertise into AI integration — including RAG pipelines, LangChain agents, and prompt engineering — positions me to contribute to modern engineering teams from day one. Available immediately for part-time, contract, or full-time roles.

Skills

Languages:	Java, Python, JavaScript
Backend:	Spring Boot, Spring Security, Django, RESTful APIs, Microservices
Frontend:	React.js, HTML5, CSS3, Bootstrap, Thymeleaf
Databases:	PostgreSQL, SQL, Relational Schema Design, ChromaDB
AI & ML:	LangChain, LLM Integration (OpenAI, Gemini, Watsonx), Prompt Engineering, RAG
Cloud & Tools:	IBM Cloud, Docker, GitHub Actions, Git, Maven, BCrypt
Concepts:	RESTful API Design, Microservices, JWT, RBAC, CI/CD, OOP
Soft Skills:	Team Leadership, Problem Solving, Effective Communication, Adaptability

Projects

StudyMate AI – Intelligent RAG-Based Study Assistant

Tech Stack: Python · LangChain · ChromaDB · Google Gemini 2.0 Flash · PyMuPDF · Docker · JWT · GitHub Actions

- Built and deployed a high-performance asynchronous RAG pipeline to process large-scale textbook PDFs and automate personalised study content generation.
- Engineered a semantic search index using ChromaDB and sentence-transformers (all-MiniLM-L6-v2) to accurately retrieve top-K relevant document chunks with zero-blocking operations.
- Leveraged Google Gemini 2.0 Flash to orchestrate multi-functional AI features: automated summary generation, interactive quiz creation, 3D flip flashcards, and dynamic day-by-day study plans.
- Implemented strict prompt engineering to force structured outputs and embed page-specific citations [Page X] via PyMuPDF, successfully mitigating LLM hallucinations.
- Hardened system security with a token-bucket rate limiter (60 req/min per IP), JWT authentication with bcrypt password hashing, and isolated multi-user data scoping.
- Containerised the dual-store architecture (SQLite + ChromaDB) with Docker Compose and configured an automated CI/CD pipeline via GitHub Actions for linting, testing, and production audits.

ApexGrid – F1 Race Ticket Booking Platform

Tech Stack: Java 17 · Spring Boot 3.2 · Spring Cloud Gateway · React 18 · TailwindCSS · PostgreSQL · JWT

- Architected a production-ready microservices application across three independent Spring Boot services (API Gateway, User Service, Race Service) with centralised routing via Spring Cloud Gateway.
- Implemented JWT-based stateless authentication with BCrypt hashing, role-based access control (USER / ADMIN), and token validation shared across all services.
- Engineered a real-time quota management system using PostgreSQL database triggers — decrementing quota on booking, restoring on cancellation, with a 30-minute auto-expiry scheduler.
- Built a full Admin Panel with race/circuit CRUD, an analytics dashboard powered by database views, and Swagger UI documentation for both services.

- Developed a premium React 18 frontend with glassmorphism UI, skeleton loaders, SVG countdown timers, toast notifications, and protected role-based routing.

Finance Management System

Tech Stack: Java · Spring Boot · Spring Security · Thymeleaf · PostgreSQL · BCrypt

- Engineered a full-stack finance web application featuring multi-role authentication (BCrypt + Spring Security), complete CRUD operations, and a real-time financial dashboard.
- Designed a normalised PostgreSQL schema and MVC backend with clean REST API endpoints, enabling seamless front-to-back data flow.

Universe Interactive Website

Tech Stack: HTML5 · CSS3 · JavaScript · Astronomy API · Animation Libraries

- Developed a space-themed interactive website featuring scroll-triggered animations, live astronomy data via REST API, and persistent dark/light mode.

Education

Associate Degree in Informatics Management	Sept 2024 – May 2027 (expected)
Universitas Nasional PASIM · Bandung, West Java GPA: 3.6 / 4.0	
Senior High School — MAN 1 Bandung Barat	Graduated 2024
Extracurricular: Robotics Club	

Experience

Lead Intern – IT & Business Operations	2025 – Present Bandung
PT Resik Cemerlang (Pasim Group)	
• Directed a 5-person intern team, assigning tasks and managing schedules to ensure all IT support tickets were resolved within agreed SLAs. • Administered weekly laptop maintenance — security patching, antivirus scanning, and hardware fault resolution — across the organisation in coordination with the Hardware Division. • Maintained and quality-checked thousands of medical device product specifications in Excel, delivering weekly operational reports to senior management.	
Lead Intern – IT Hardware	2024 Bandung Barat
SMP Negeri 2 Batujajar	
• Led full hardware maintenance cycle for a computer lab and administrative network: fault diagnosis, component repair, peripheral troubleshooting, and printer servicing. • Established a hardware inventory and maintenance log system in coordination with school administration, reducing repeat-fault response time.	

Certifications

Python for Everybody Specialization — University of Michigan	2026
Programming for Data Science & Entry-Level Python Programmer — HarvardX	2025
Introduction to Cloud Computing — IBM	2026

Languages & Availability

Indonesian — Native English — Professional Working	Available immediately (Part-time / Contract / Full-time)
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